POLITECNICO MILANO 1863 Academic year 2024/2025	Course RF SYSTEMS	**Student ID or Persona code** _____
	Student Signature _____	**Student Name, Surname** _____

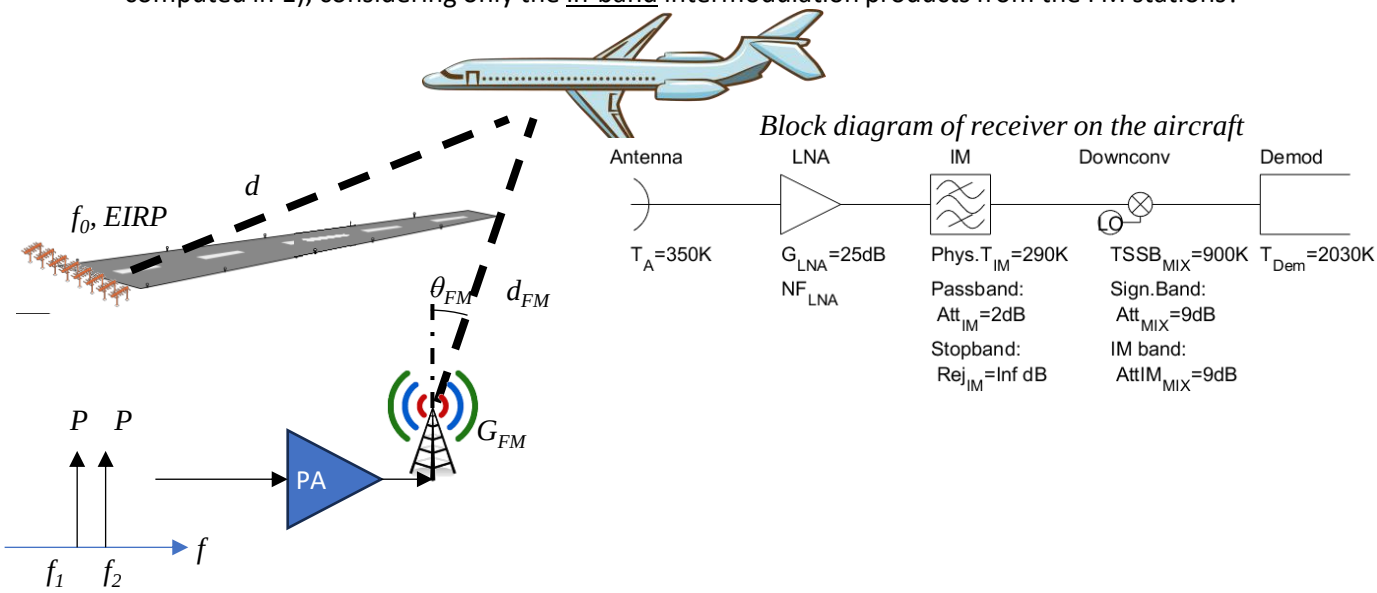
15 January 2025 Full (26 pts available; ≥15 needed)

Please answer the following questions/problems, providing a brief explanation in English language of the steps/computations involved. Please specify units for all numeric results requiring them. Allowed support material: textbooks, notes, scientific calculator, laptop/tablet (offline).

Exercise 1 (13 points)

A receiver of an Instrument Landing System (ILS, shown in the figure below) at the frequency f_0=110.1MHz on an aircraft has sensitivity S=-80dBm (measured at its antenna port) to guarantee SNR=37dB due to the total system noise of the receiver. The aircraft is equipped with an isotropic antenna with efficiency η=0.6 and the signal bandwidth filtered by the demodulator is B=111kHz. The ILS transmitter has EIRP=46dBm, and it is perfectly aligned toward the aircraft.

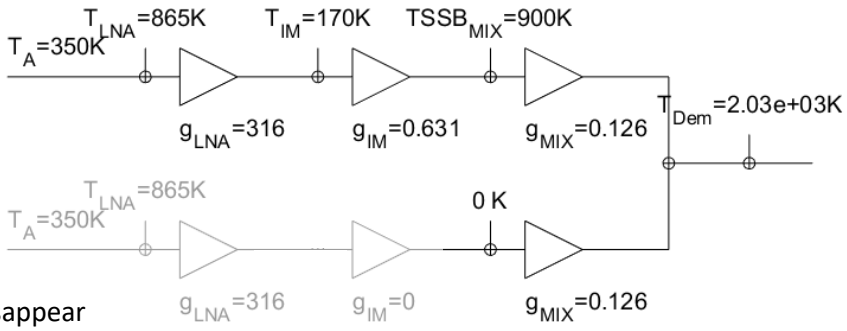
a) What is the max. distance *d* from the transmitter to have SNR=37dB (due to the noise of the rx)?
b) Draw the noise diagram, write T_{sys} and compute the needed NF_{LNA} (everything matched at 50Ω).
c) A radio tower handles two FM stations at f_1=102.5MHz and at f_2=106.3MHz (a sinusoid each at their carrier frequency) each with power P=31dBm. These are then amplified by a PA (G=25dB, IP3=84dBm) and radiated. Draw the spectrum at the antenna port of the tower, with frequencies and avg power of each tone (pure 3rd order model; neglect compr. of main tones).
d) Using the results of c), the antenna of the FM tower has max gain G_{FM}=-1dBi and directivity function f(θ,ϕ)=sin(θ), with θ=0 pointing directly toward the sky and θ=90° pointing toward the horizon (maximum radiation direction). The aircraft is at θ_{FM}=0.5° and distance d_{FM}=10km. What is the equivalent SNIR=SIGNAL/(THERM.NOISE + INTERMOD) at the aircraft at the distance computed in 1), considering only the <u>in-band</u> intermodulation products from the FM stations?



Solution

a) The sensitivity is the lowest received power level which guarantees the requested performance (in this case SNR=37dB due to the system noise of the receiver). Thus, we need a link budget which guarantees a received signal of -80dBm: S=PRx=PTx+GTx-FSL+GRx; with PTx+GTx=EIRP=46dBm, GRx=10Log(1 · 0.6)=-2.2dBi (the directivity of the isotropic antenna is 0dBi, and it has efficiency=0.6). Thus FSL=EIRP-S+GRx=123.8dB=20Log(4 π d/λ)→ d=336km.

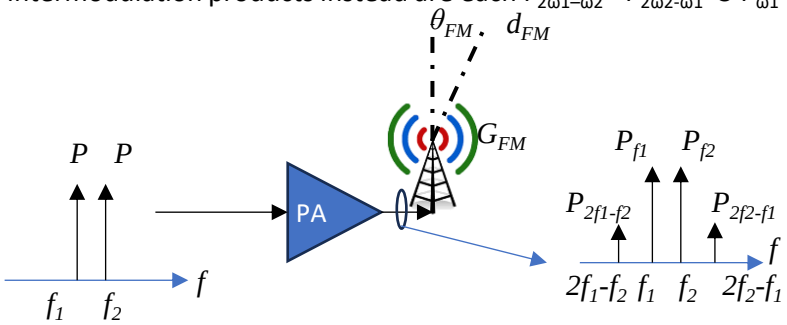
b) The noise diagram is shown on the right.
TLNA is however unknown, so far.
But SNR=S/Pneq
→Pneq=-117dBm=0.002pW=$kT_{sys}B$
→ Tsys=1300K at the ant. port.
The filter does not introduce thermal noise in its stopband.
The IM filter has also infinite rejection in the IM band, thus g_{IM}=0 there and all noise contrib. before the IM filter in the IM band disappear (shown shaded, in gray).
The expression of T_{sys} is:



$$\left(\frac{T_A}{350K}\right)+\left(\frac{T_{LNA}}{}\right)+\left(\frac{T_{IM}}{170K}\right)/\left(\left(\frac{g_{LNA}}{316}\right)\right)+\left(\frac{TSSB_{MIX}}{900K}\right)/\left(\left(\frac{g_{LNA}}{316}\right)\cdot\left(\frac{g_{IM}}{0.631}\right)\right)+\left(\left(\left(\left(\frac{T_A}{350K}\right)+\left(\frac{T_{LNA}}{}\right)\right)\cdot\left(\frac{g_{LNA}}{316}\right)\right)\cdot\left(\frac{g_{IM}}{0}\right)\right)\cdot\left(\frac{g_{MIX}}{0.126}\right)/\left(\frac{g_{MIX}}{0.126}\right)/\left(\left(\frac{g_{LNA}}{316}\right)\cdot\left(\frac{g_{IM}}{0.631}\right)\right)+\left(\frac{T_{Dem}}{2030K}\right)/\left(\left(\frac{g_{LNA}}{316}\right)\cdot\left(\frac{g_{IM}}{0.631}\right)\cdot\left(\frac{g_{MIX}}{0.126}\right)\right)$$

Thus T_{sys}=1300K=350K+T_{LNA}+85K → T_{LNA}=865K→NF_{LNA}=(1+T_{LNA}/290K)=3.98=6dB

c) The FM radio tower amplifies two signals, modeled as a sinusoid each, with same power P=31dBm. Thus, it amplifies a 2-tone signal with average power 34dBm at the input of the PA.
The spectrum at the output will contain the main tones (f1=102.5MHz, f2=106.3MHz) amplified, and their intermodulation products (2f1-f2=98.7MHz, 2f2-f1=110.1MHz) created by the non-linear behavior of the PA.
The main tones are amplified by G=25dB (since we are neglecting their compression), and thus each line at the antenna port is $P_{ω1}$=$P_{ω2}$=56dBm (the combined average power of this 2-tone signal at the output is Pm=59dBm).
The intermodulation products instead are each $P_{2ω1-ω2}$= $P_{2ω2-ω1}$=3·$P_{ω1}$-2·IP3=0dBm.



d) The upper intermodulation product coincides exactly with the ILS signal, at 110.1MHz, thus it acts as an interferer toward the landing aircraft. Instead, the main FM stations and the lower intermodulation products are filtered by the receiver circuitry. The upper intermodulation product hence reaches the aircraft antenna port following the link budget equation, in dB, with a penalty term due to the misalignment of the aircraft with respect to the maximum radiation direction of the FM transmitter (the aircraft is at $θ_{FM}$=0.5°, while the max. radiation is at θ=90°, where f(θ,ϕ)=1):
$P_{2ω2-ω1@aircraft}$=$P_{2ω2-ω1}$+G_{FM}+10Log(sin(0.5°))-FSL_{FM}+G_{RX}=-117dBm, with FSL=20Log(4πd_{FM}/λ)=-93dB
Thus the undesired intermodulation product reaches the aircraft's antenna port with the same power as the equivalent thermal noise computed in b). The useful signal, computed in a), is still equal to the sensitivity S=-80dBm, since the distance from the ILS transmitter has not changed. Hence the equivalent SNIR, including the equivalent thermal noise and the interfering intermod. product will be halved (34dBm): SNIR=$10^{(-90/10)}$mW/($10^{(-117/10)}$mW + $10^{(-117/10)}$mW)=2505=34dB.

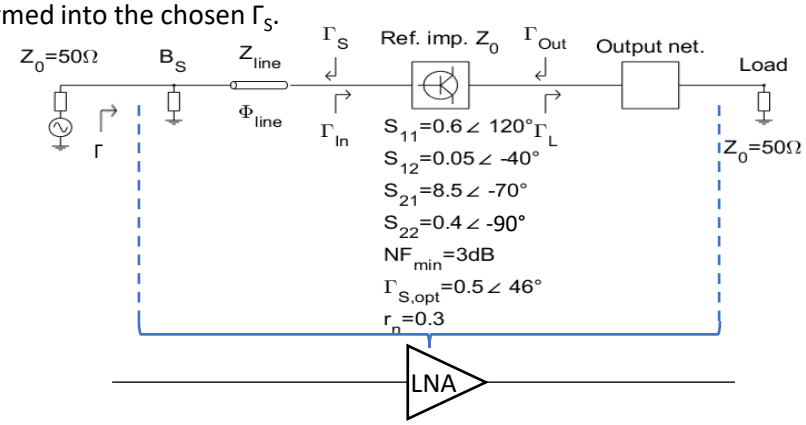
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Exercise 2 (13 points)

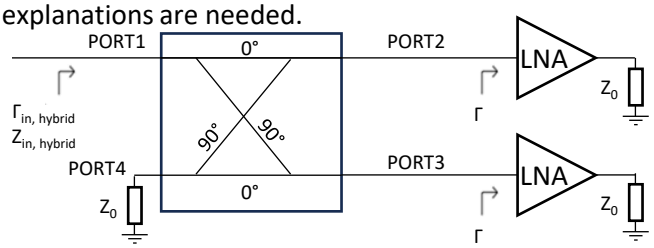
An HBT, once biased, exhibits the scattering parameters shown below, referred to 50Ohm at the frequency of 6.5GHz. It must be used to design an LNA with maximum transducer gain while guaranteeing NF=3.9dB. The output network is lossless and does not impose any constraint.

a) Provide a reasonable choice for Γ_S and Γ_L, verifying stability and write the obtained transducer gain (approximate at 1 decimal digit)?

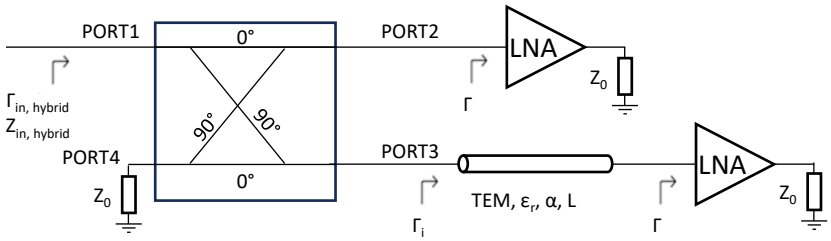
b) Design the input network (choose Z_{line} and compute ϕ_{line}, B_S in Siemens) so that an external 50 Ω is transformed into the chosen Γ_S.



c) When two identical amplifiers, as designed in a) and b) are fed by means of an ideal lossless 90° 3dB hybrid designed at 50Ohm, (hint: recall the balanced amplifier) is the input impedance observed from the input port of the hybrid $Z_{in,hybrid}$ equal to 50Ohm? Computations are not strictly required, but explanations are needed.



d) Similarly to c), but an additional lossless (α=0dB/m) TEM line with ε_r=4.4 and physical length L=11mm is inserted before the second amplifier: is the input impedance observed from the input port of the hybrid $Z_{in,hybrid}$ equal to 50Ohm? Computations are required only to compute the electrical length of the additional line, but explanations are necessary for the whole answer.

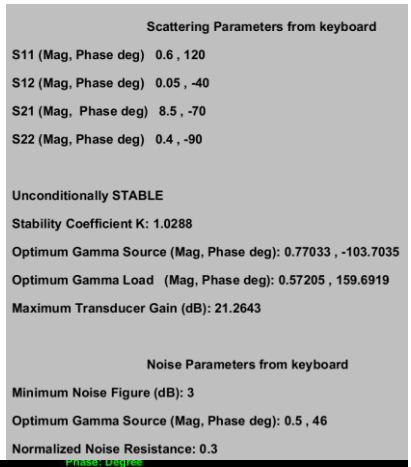


Solution

a) The S parameters of the device are written in the electronic Smith Chart, which shows that the device is unconditionally stable, with maximum transducer gain 21.26dB. However, in this case the objective is to enforce the required NF (3.9dB), and then maximize the transducer gain. To do this, the NF=3.9dB circle is drawn, which is the locus of points where Γ_S should be selected (NF depends on Γ_S, not on Γ_L).

To maximize the transducer gain, hence, we look for the maximum available gain G_a (since it also depends on Γ_S, not on Γ_L) achievable while remaining on the NF=3.9dB circle. Several G_a circles are drawn, until the one with G_a=20.5dB is found to be tangent with the NF=3.9dB circle. The tangent point occurs at Γ_S=0.353 ∠-96° (click on this point to make it the Current point).

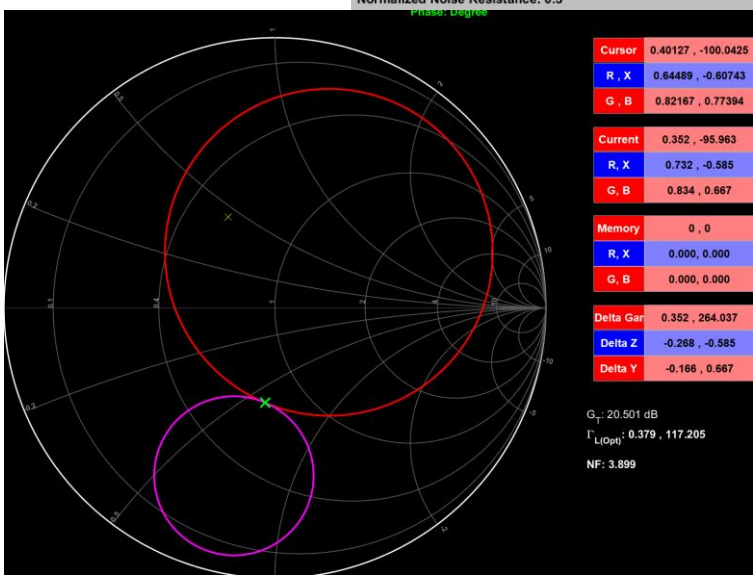
This point is acceptable for stability, since the device is unconditionally stable, hence the stability region for the source is the whole complex plane.



To make $G_T=G_a$=20.5dB, we need to impose conjugate matching at the output. Thus we compute the corresponding optimum Γ_L for our choice of Γ_S, which is Γ_L=0.379 ∠117.2°, giving NF=3.9dB and G_T maximized at 20.5dB.

This value is also acceptable for stability since the device is unconditionally stable. Any other passive choice of Γ_L would still be acceptable for stability, but G_T would be less than G_a (not maximum).

The amplifier is thus matched at its output, although not matched at its input (Γ≠0).

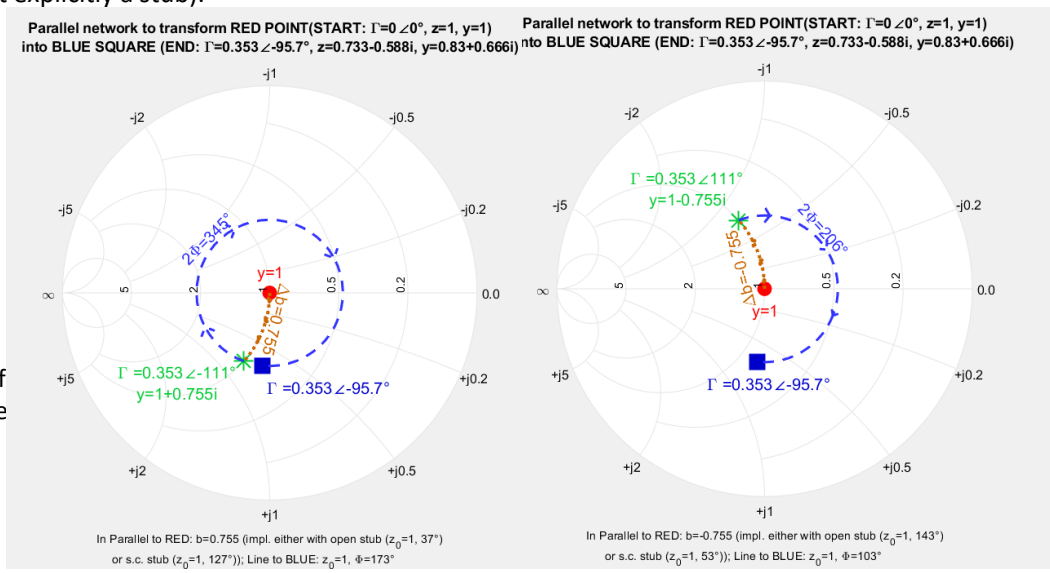


b) The input net must transform the external 50 Ω source into Γ_S=0.401 ∠-100° (referred to 50 Ω) required by the amplifier. To avoid any renormalization and guarantee that the input network is realizable, we choose Z_{line}=50Ω.

Thus the starting point of the transformation is Γ_{start}=0 (red, center of the Smith chart) and the desired ending point is Γ_S (blue). The design of the input network follows the typical single stub in parallel (although b_S is not explicitly a stub).

The first solution (left) requires normalized b_S=0.755, hence B_S=0.015S, and ϕ_{line}=173°.

The second solution (right) requires b_S=-0.755, hence B_S=-0.0151S and ϕ_{line}=103°.



c) When fed from port 1, two identical amplifiers present the same reflection Γ at the Through and Coupled ports of the hybrid (2 and 3). Thus, the reflected waves from them (same amplitude, phase difference 90°) re-enter the hybrid and cancel out at the port1 (180° phase difference).

(the reflected waves in fact add constructively only the matched load on port 4). Thus the the input port of the hybrid does not observe any reflected wave, regardless of the Γ at the ports. Thus $\Gamma_{in\ hybrid}$=0, thus $Z_{in,hybrid}$=50Ohm.

d) The additional line has an electrical length $\phi=\beta L=(\omega/c)\cdot sqrt(\varepsilon_r)\cdot L=\pi$ rad=180°→ Half wavelength line. Therefore, regardless of the characteristic impedance of the line, ideally, whatever reflection coefficient Γ terminates the other end, it will be replicated at the input: $\Gamma_i=\Gamma\, e^{-2j\phi}=\Gamma$. Thus the hybrid is loaded by the exact same conditions as in point c): both ports are loaded by Γ, and the input impedance observed at port 1 is hence again $Z_{in,hybrid}$=50Ohm.

This is true only at a single frequency, i.e. where the line is exactly 180° long (or multiples), as in this case.